

Task 3 :- Numbers :- From Decimal to Hexadecimal

Human beings have 10 fingers, it is most plausible explanation for why we use decimal (base-10) system.

Computers, on other hand, have 2 fingers states that they understand :-

0 and 1
* low and High in Voltage range (example: transistor-transistor logic)

* North and South in magnetic polarity (Ex:- hard disk drives)

* light presence (Ex:- Fiber optics)

In previous task, we saw that we can use set of bits to fine tune the steps. For instance, 1 bit can represent 2 states, while 8 bits can represent 256 states. Let's learn math beyond it.

* Simple math terms, A number such 213 is actually same as saying $200 + 10 + 3$. Is it this right?

* In more technical 213 can written as $2 \times 10^2 + 1 \times 10^1 + 3 \times 10^0$. As would remember from math class $10^2 = 100$, $10^1 = 10$, $10^0 = 1$.

Consequently, this was just formal mathematical way of saying $= 2 \times 100 + 1 \times 10 + 3 \times 1$.

Binary Numbers :-

The binary (base 2) system can be expressed similarly to what we wrote earlier, discussing decimal (base-10) system. The key difference is binary limited to two digits, 0 and 1 everything is power of 2.

* Let's consider couple of examples.

Ex:- binary number 1001 can be expressed as $1001 = 1 \times 2^3 + 0 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 = 1 \times 8 + 0 \times 4 + 0 \times 2 + 1 \times 1$

$= 8 + 0 + 0 + 1 = 9$. We just demonstrated how to write 9 in binary.

* Following same approach, it won't be challenging to convert binary numbers 0000, 0001, 0010, 0011 to decimal system. Let's go through 4 conversions

$$0000 = 0 \times 2^3 + 0 \times 2^2 + 0 \times 2^1 + 0 \times 2^0 = 0 \quad | \quad 0001 = 0 \times 2^3 + 0 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 = 1 \times 1 = 1$$

$$0010 = 0 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 0 \times 2^0 = 2 \quad | \quad 0011 = 0 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0 = 2 + 1 = 3$$

If this is your first time working with different bases, you might have already figured out right most

digit is multiplied by 2^0 in case of binary (base 2) system and multiplied by 10^0 in case of

decimal (base-10) system. Then next digit is multiplied by 2^1 in base 2 system and by 10^1 in base

-10 system.

Exercise:- we will convert 4 more binary numbers, 1100, 1101, 1110 and 1111 to decimal system.

$$1100 = 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 0 \times 2^0 = 8 + 4 = 12 \quad | \quad 1101 = 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 = 8 + 4 + 1 = 13$$

$$1110 = 1 \times 2^3 + 1 \times 2^2 + 1 \times 2^1 + 0 \times 2^0 = 8 + 4 + 2 = 14 \quad | \quad 1111 = 1 \times 2^3 + 1 \times 2^2 + 1 \times 2^1 + 1 \times 2^0 = 8 + 4 + 2 + 1 = 15$$

Having come for, it's time to revisit hexadecimal system

1 = light on 0 = light off

In task 2, we grouped every 4 bits into a single hexadecimal digit. As we seen earlier

hexadecimal digit ranges between 0 and F. By taking closer look at table below, you will notice

a hexadecimal digit ranges b/w 0 and 15 in decimal system. To replace 10, 11, 12, 13, 14 and 15

each with single letter/digit A, B, C, D, E and F are chosen. The table of Hexadecimal

Decimal Number	Hexadecimal digit	Binary representation
0	0	0000
1	1	0001
2	2	0010
3	3	0011
4	4	0100
5	5	0101
6	6	0110
7	7	0111
8	8	1000
9	9	1001
10	A	1010
11	B	1011
12	C	1100
13	D	1101
14	E	1110
15	F	1111

lets say we want to convert hexadecimal Number 9BDF to decimal.

$$9BDF = 9 \times 16^3 + 11 \times 16^2 + 13 \times 16^1 + 15 \times 16^0 = 9 \times 4096 + 11 \times 256 + 13 \times 16 + 15 \times 1 = 39,903$$

optional:- octal numbers

The Octal system refers to base 8. In other words we use digits b/w 0 and 7.

while hexadecimal uses base 16 and groups 4 bits, Octal system uses base 8 and groups 3 bits.

Octal digit to binary counterparts.

Decimal Number	Octal Digit	Binary Representation
0	0	000
1	1	001
2	2	010
3	3	011
4	4	100
5	5	101
6	6	110
7	7	111

optional covering from octal to Decimal System :-

$$\text{Ex: } - 357 = 3 \times 8^2 + 5 \times 8^1 + 7 \times 8^0 = 3 \times 64 + 5 \times 8 + 7 \times 1 = 239$$

Quiz:-

1) What is Hexadecimal FF in binary?

$$15 \times 16^1 + 15 \times 16^0 = 15 \times 16 + 15 \times 1 = 255 \quad \rightarrow \text{Decimal representation}$$

$$\text{Binary} = 1111 \ 1111 = F = 1111_{16}$$

2) What is hexadecimal AB in decimal?

$$\text{A) } A=10 \ B=11 \quad 10 \times 16^1 + 11 \times 16^0 = 160 + 11 = 171$$

3) Convert hexadecimal FFFF FF to decimal. After you round up decimal value to nearest million, how many million is that?

$$\text{A) } 16,777,215 = 17 \text{ million} \quad \rightarrow \text{million}$$

Conclusion we covered

* Decimal (Base 10) System * Binary (Base 2) System * Hexadecimal (Base 16) System * Octal (Base 8) System.

Bit $\rightarrow$ It's short for binary digit, it can be either 0 or 1

Byte :- one million Systems, byte is 8 bits. It can be also referred as octet.

Hex color :- A color is represented as combination of red, green and blue. If one byte is assigned for each of primary colors (red, green and blue), we can get 16 million color combinations.

